Grade 8
Unit 2
Lesson 5 Practice

Name__Answer Key________
Date________________________
Period_______________________

For each of the values below in questions 1-5, identify whether it is rational or irrational.

1. $\pi + 5.5$ is a ○ rational. ○ irrational. number because it ○ can ○ cannot be expressed as a ratio of two integers.

2. $-\sqrt{135}$ is a ○ rational. ○ irrational. number because it ○ can ○ cannot be expressed as a ratio of two integers.

3. $\sqrt[3]{-512}$ is a ○ rational. ○ irrational. number because it ○ can ○ cannot be expressed as a ratio of two integers.

4. $-12.12\overline{3}$ is a ○ rational. ○ irrational. number because it ○ can ○ cannot be expressed as a ratio of two integers.

5. $0.5435432\ldots$ is a ○ rational. ○ irrational. number because it ○ can ○ cannot be expressed as a ratio of two integers.

Consider the expression $\sqrt{150}$.

6. Between which two whole numbers does $\sqrt{150}$ lie?
$\sqrt{144} = 12$
$\sqrt{169} = 13$
$\sqrt{150}$ lies between 12 and 13.

7. To the nearest tenth, approximate the value of $\sqrt{150}$. Then plot the approximate location on the number line below.

8. Use the approximation of $\sqrt{150}$ to approximate the value of each expression in the table.

Expression	Approximate Value
$12-\sqrt{150}$	$12-12.2 \approx -0.2$
$2\sqrt{150}$	$2(12.2) \approx 24.4$
$\frac{\sqrt{150}}{6}$	$\frac{12.2}{6} \approx 2$

9. Approximate the value of each expression.

	Expression	Approximate Value
A	$\sqrt[3]{-198}$	$\sqrt[3]{125} = -5$ $\sqrt[3]{-198} = -5.8$ $\sqrt[3]{-216} = -6$
B	$\frac{1}{2}\sqrt[3]{-198}$	$\frac{1}{2}(-5.8) \approx -2.9$
C	$-\frac{3}{4}\pi$	$-\frac{3}{4}(3.14) \approx -2.3$
D	$\pi - 7$	$3.14-7 \approx -4$

10. Plot and label the approximate location of each expression from question 9 using letters A, B, C, and D on the number line.